Lab 3, Act 3

From counts to a continuous measurement

NRES 746

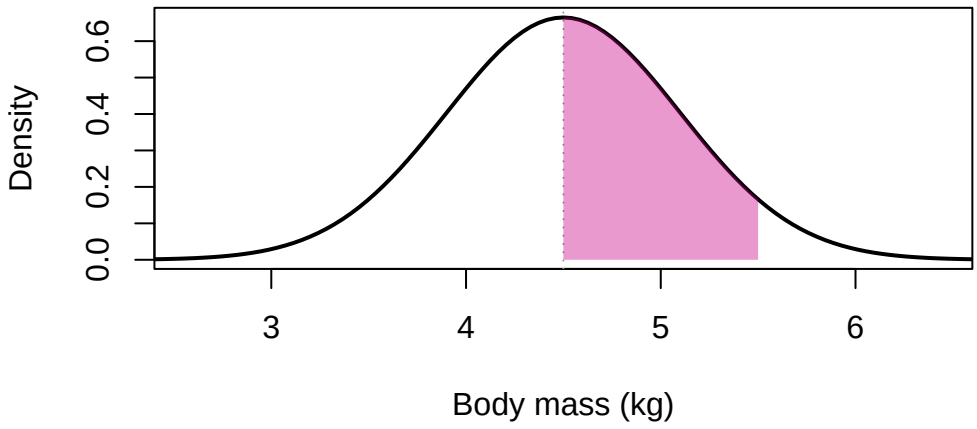
Answer key. Answers are in magenta.

Picking up where we left off

So far this lab has treated detections as discrete counts (0, 1, 2, … detections). But some of the quantities the field team records are continuous. Suppose adult body mass for this carnivore, estimated from reference objects visible in camera-trap images, is well described by a normal distribution with mean $\mu = 4.5$ kg and standard deviation $\sigma = 0.6$ kg.

Part 1: Reading a density curve (individual, ~10 min)

Below is the probability density function (pdf) for this normal distribution.



1a. Shade it by hand. On the curve above, shade the region that represents $P(4.5 < \text{mass} < 5.5)$.

1b. Integral notation. Given that the probability density function can be written $f(x)$ where x is body mass and f(x) is the normal (Gaussian) density function (you can ignore the specific mathematical equation for the Gaussian density function for now), please write out the same region described above as a definite integral.

$P(4.5 < \text{mass} < 5.5) = \int_{4.5}^{5.5} f(x)\,dx$ (shaded in magenta above)

1c. What is $P(\text{mass} = 4.5)$ exactly? Why – what's different about a single point on a continuous curve versus an interval?

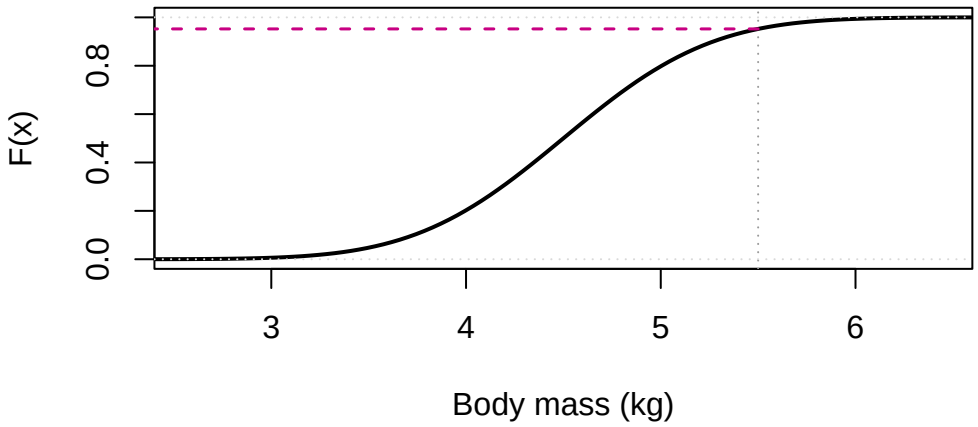
0. A single point has no width, so there's no area under the curve above it; probability builds up only over an interval.

1d. The height of the curve at mass = 4.5 is not zero (in fact, it's the highest point on the whole curve). If that height isn't a probability, what does it actually represent? HINT: the technical answer is a 'probability density'. Try to write out in words what a 'probability density' means.

Probability per unit of mass: how concentrated probability is near 4.5 kg. Height $\times$ a small width $\approx$ the probability of that small interval (e.g. $0.665 \times 0.1 \approx 0.07$ for 4.45 to 4.55 kg).

Part 2: Reading a CDF (individual, ~5-8 min)

Below is the cumulative distribution function (CDF), $F(x) = P(\text{mass} \leq x)$, for the same distribution.



2a. Using the CDF plot, estimate $P(\text{mass} > 5.5)$. Record your estimate – you don't need to be precise, just in the right neighborhood.

About 0.05: $F(5.5) \approx 0.95$ (magenta line), and $P(\text{mass} > 5.5) = 1 - F(5.5)$. Anything from about 0.03 to 0.08 is fine.

Part 3: Standardize and look it up (individual, ~10-12 min)

Graph-reading only gets you so far. The **standard normal table** (Table 1, at the end of this Act) gives exact values of $\Phi(z) = P(Z \leq z)$ for the *standard* normal distribution ($\mu = 0$, $\sigma = 1$). To use it for a normal distribution with a different mean and standard deviation, first convert your value of interest to a **z-score**:

$$z = \frac{x - \mu}{\sigma}$$

This rescales x into "number of standard deviations away from the mean," (standardized difference from the mean) which is exactly what the table expects.

3a. Compute the z-score for $x = 5.5$ kg.

$z = (5.5 - 4.5)/0.6 \approx 1.67$

3b. Look up $\Phi(z)$ for the z-score you just computed (round z to the nearest 0.01 if needed), then use it to compute $P(\text{mass} > 5.5)$. How close was this to your graph-reading estimate from 2a?

$\Phi(1.67) = 0.9525$, so $P(\text{mass} > 5.5) = 0.0475$ (about 4.7%), close to the 2a estimate.

3c. Now go the other direction. Search the table for the value closest to 0.90, and read off the z-score (90% quantile). This is (approximately) the z-score marking the heaviest 10% of the population (only 10% of individuals expected to exceed this standardized value). In the space below, convert that z-score back into a body mass.

Closest table value: $\Phi(1.28) = 0.8997$, so $z \approx 1.28$ and mass $= 4.5 + 1.28 \times 0.6 \approx 5.27$ kg.

Group discussion and revision (~5 min)

Compare your Part 3 answers with your group. One thing worth discussing: Would this z-table give the right answer if you standardized Act 2's binomial detection counts the same way?

Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion ("muddiest points") or topics you want to review or dust off.

No, not reliably. Standardizing changes location and scale, not shape; it gives a standard normal only if the variable was normal to begin with. Act 2's counts are discrete and right-skewed.

Table 1: Standard normal cumulative probabilities, $\Phi(z) = P(Z \leq z)$

Row = z to one decimal place; column = second decimal place. Example: for $z = 1.67$, read the row "1.6" and the column ".07".

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998